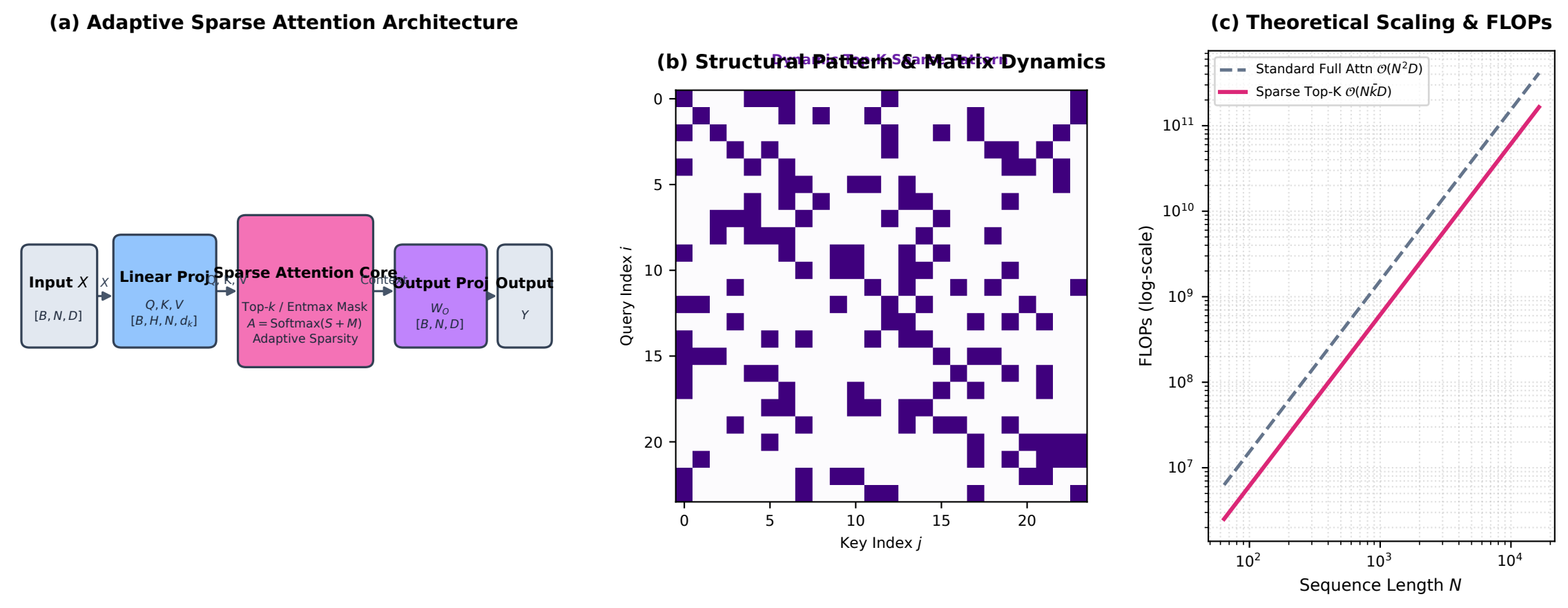
(a) Adaptive Sparse Attention Architecture
Input X
$[B, N, D]$
X
Linear Proj
Q, K, V
$[B, H, N, d_k]$
Q, K, V
Sparse Attention Core
Top-k / Entmax Mask
$A = \text{Softmax}(S + M)$
Adaptive Sparsity
Context
Output Proj
W_O
$[B, N, D]$
Output
Y
(b) Structural Pattern & Matrix Dynamics
Dynamic Top-K Sparse Pattern
Query Index i
Key Index j
0
5
10
15
20
(c) Theoretical Scaling & FLOPs
Standard Full Attn $\mathcal{O}(N^2D)$
Sparse Top-K $\mathcal{O}(N\bar{k}D)$
FLOPs (log-scale)
Sequence Length N
10^7
10^8
10^9
10^{10}
10^{11}
10^2
10^3
10^4